



NEWS CREDIBILITY ANALYSER

NLP-Based Fake News Detection
using Machine Learning

Presented By
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Problem Statement

Problem

The rapid spread of fake news through online platforms makes it difficult for users to distinguish credible information from misinformation.

Challenges

- Large volume of news generated daily
- Manual fact-checking is time-consuming
- Misinformation spreads quickly
- Need for automated credibility assessment

Proposed Solution

An NLP-powered system that classifies news articles as:

- Credible News
- Fake News



Objectives

- Build a fake news classification system
- Apply Natural Language Processing techniques
- Compare multiple machine learning models
- Provide confidence-based predictions
- Explain model decisions through feature analysis
- Deploy the solution using Streamlit



Dataset

ISOT Fake News Dataset

Category: Credible News - 21,191

Category: Fake News - 17,448

Total Articles - 38,639

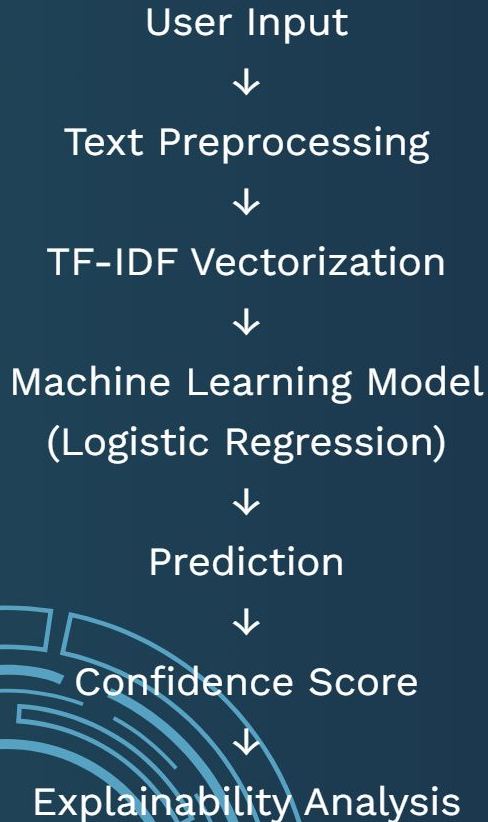
Dataset Features

- Title
- News Content
- Label

Data Cleaning

- Remove duplicates
- Remove missing values
- Text normalization
- Stopword removal
- Lemmatization

System Architecture



Methodology

Text Preprocessing

- Lowercasing
- URL Removal
- HTML Removal
- Reuters Leakage Removal
- Stopword Removal
- Lemmatization

Feature Engineering

TF-IDF Vectorization

- Unigrams
- Bigrams
- 30,000 Features



Model Training

Model Selection

Chosen Model:

Logistic Regression

Reasons:

- High accuracy
 - Fast inference
 - Explainable predictions
- 

Models Evaluated & their purpose

Logistic Regression - Final Model

Multinomial Naive Bayes - Baseline

Random Forest - Comparison



Results

Metric	Score
Accuracy	98.64%
Precision	98.69%
Recall	98.57%
F1 Score	98.63%
ROC-AUC	99.86%

Observation

The model achieved excellent classification performance while maintaining interpretability.



Explainable AI



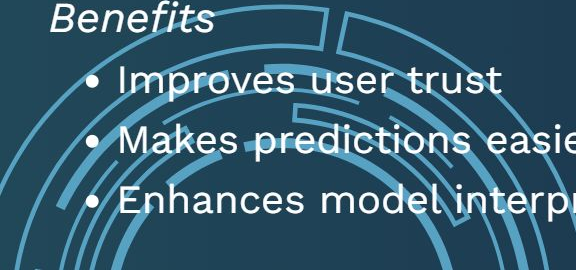
Explainable Predictions

Unlike traditional classifiers, our system provides insights into why a news article is classified as Credible or Fake.

Features

- News Classification
- Confidence Score
- Important Keyword Analysis
- Transparent Decision Making

Benefits



- Improves user trust
- Makes predictions easier to understand
- Enhances model interpretability

Key Takeaway:

The system not only predicts the credibility of a news article but also explains the factors influencing the prediction.

Application Demo

NEWSCREDIBLE

HOME ANALYSE NEWS MODEL INSIGHTS ABOUT

Analyse News Credibility Using NLP & Machine Learning

GO

An NLP classifier that reads any news article and tells you whether it looks credible or fabricated
— trained on 38,639 real-world articles, built end-to-end with TF-IDF and Logistic Regression.

38.6k+

Articles in Training Set

3

Models Compared

99.04%

Final Model Accuracy

30k

TF-IDF Features

ANALYSIS RESULTS



Confidence Level — 96.6%



WHY THIS PREDICTION

Terms found in this article that most influenced the model — drawn directly from the trained Logistic Regression coefficients.

Pushed toward Fake

from news received voted full washington bill member

Pushed toward Credible

said wednesday said statement statement month senate

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Check an article's credibility

Paste a headline and body below, or load a sample to see the model in action.

Try a Fake-style Sample

Try a Credible-style Sample

ARTICLE INPUT

Article Title

Senate Committee Advances Infrastructure Funding Bill After Bipartisan Vote

Article Body

WASHINGTON (Reuters) - The Senate Appropriations Committee voted 21-9 on Wednesday to advance a \$1.2 billion infrastructure funding bill, sending the measure to the full chamber for consideration. The bill, which allocates funds for bridge repairs and public transit upgrades across 14 states, received support from members of both parties following weeks of negotiation. Committee chairwoman said in a statement that the bill reflects a compromise reached after input from state transportation officials. The legislation is expected to reach the Senate floor for a full vote next month.

TIPS FOR A RELIABLE RESULT

- Paste the real headline, not a paraphrase
- Include the full article body — at least a few paragraphs
- English-language political / news-style text works best
- Very short snippets often land in "Uncertain"

HIGHLIGHTED ARTICLE

The same terms shown as chips above, highlighted directly in your article — so you can see exactly where each signal came from.

Senate Committee Advances Infrastructure Funding **Bill** After Bipartisan Vote

WASHINGTON (Reuters) - The **Senate** Appropriations Committee **voted** 21-9 on **Wednesday** to advance a \$1.2 billion infrastructure funding **bill**, sending the **measure** to the **full** chamber for consideration. The **bill**, which allocates funds for bridge repairs and public transit upgrades **across** 14 states, **received** support from **members** of both **parties** following weeks of negotiation. Committee chairwoman **said** in a **statement** that the **bill** reflects a compromise reached after input from state transportation officials. The legislation is expected to reach the **Senate** floor for a **full** vote next **month**.

Pushed toward Fake Pushed toward Credible

DOWNLOAD REPORT

Save this verdict, confidence score, and contributing terms as atext file — useful for attaching to an assignment or portfolio write-up.

Download Analysis Report (.txt)

PREDICTION HISTORY (THIS SESSION)



Limitations & Challenges

Limitations

- Does not verify facts in real time
- Depends on training data quality
- Cannot detect newly emerging misinformation
- Works on writing patterns rather than actual truth

Key Learning

A dataset leakage issue was discovered and removed during development, improving model reliability.



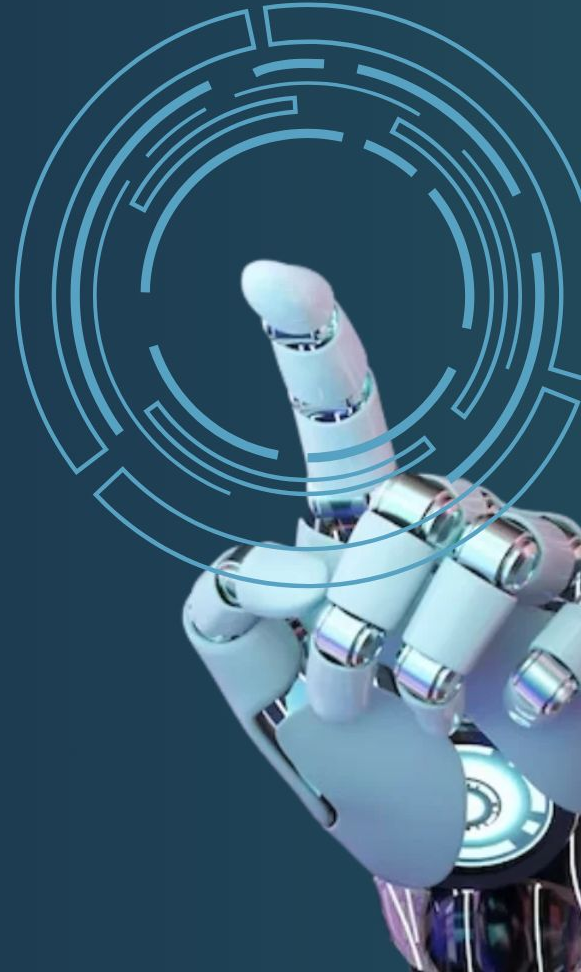
Future Scope & Conclusion

Future Scope

- Transformer-based models (BERT)
- Real-time news verification
- Source credibility scoring
- Multilingual support
- Live web integration

Conclusion

The News Credibility Analyser successfully combines NLP and Machine Learning to classify news articles with high accuracy while providing transparent and explainable predictions.



Thank You

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